(T) Roll No. .....

PAPER ID—16014

B.Tech. EXAMINATION, 2024

(Third Semester)

ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

Advanced Data Structure

Time : 3 *Hours* *Maximum Marks* : 70

Before answering the question-paper candidates should ensure that they have been supplied to correct and complete question-paper. No complaint, in this regard, will be entertained after the examination.

Note : Attempt *Five* questions in all, selecting *one* question from each Unit. Q. No. 1 is compulsory. All questions carry equal marks.

1. (a) Discuss the challenges of file fragmentation and its impact.

(b) What is the concept of files and streams in programming ?

(c) How are B+ trees used in database indexing ?

(d) What is a splay tree ? Explain the concept of splaying.

(e) Compare 2-3 trees with AVL trees and red-black trees.

(f) What is Rehashing ? When and why is it necessary ?

(g) Discuss Dictionary significance in data structures. **14**

Unit I

2. Explain, how to traverse a singly linked list ? Write an algorithm and analyze its time

complexity. Describe the process of inserting a new node in a singly linked list at the beginning, middle, and end. Illustrate with examples. 14

3. What is Hashing ? Explain, how a hash function works. Discuss the desirable properties of a good hash function with examples. Discuss collision resolution techniques in hashing. Compare separate chaining and open addressing methods. 14

Unit II

4. Explain the structure of binary trees and binary search trees (BSTs). Compare their differences with examples. Describe the traversal methods of binary trees (in-order, pre-order, post-order, and level-order). Provide algorithms for each traversal technique. 14

5. What is an AVL tree ? Discuss its significance and how it differs from a binary search tree. Explain the concept of rotations in AVL trees. Discuss the four types : LL (Left-Left), RR (Right-Right), LR (Left-Right), and RL (Right-Left) with examples. **14**

Unit III

6. Explain the concept of a set and its representation in computer science. Describe the union, intersection, and difference operations on sets. Provide algorithms for implementing these operations **14**

7. What is file organization ? Explain the importance of efficient file organization in data storage and retrieval. Describe sequential file organization. How is it implemented and what are its advantages and limitations ? **14**

Unit IV

8. What is a Graph ? Explain its components (vertices, edges, degree, etc.) with examples of undirected and directed graphs. Compare adjacency matrix and adjacency list representations of graphs. Discuss their advantages and disadvantages in terms of memory usage and performance. Explain the concept of weighted and unweighted graphs. How are weights represented in adjacency matrices and lists ? **14**

9. What is the shortest path problem in graphs ? Discuss its importance and real-world applications. Explain Dijkstra's algorithm for finding the shortest path in a graph with non-negative weights. Write the algorithm and analyze its time complexity. **14**